



WhatsApp Bot

Complete Setup Guide

A step-by-step dummies guide
from zero to a live bot on WhatsApp

Covers: Local Setup · Gmail · Railway Cloud · Product Management
July 2026

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1. What this bot does

This is a WhatsApp shopping bot. Customers can message it on WhatsApp and:

- Type `hello` or `menu` to see your product list
- Type `buy 1 2` to add 2 of product ID 1 to their basket
- Type `checkout` to see their order total

As the admin, you can send a new CSV file to the bot on WhatsApp to instantly update the product catalog.

2. How it works with WhatsApp (Meta)

What is Meta?

Meta is the company that owns WhatsApp, Facebook, and Instagram. WhatsApp runs on Meta's servers.

This bot connects to WhatsApp by pretending to be a WhatsApp Web browser session — the same way you use WhatsApp on a laptop. This means:

- You use a **real WhatsApp phone number** (personal or WhatsApp Business)
- You scan a QR code once to link the bot to that number
- The bot then sends and receives WhatsApp messages on behalf of that number
- You do **not** need to pay Meta or sign up for the Meta Business API

Important: While the bot is running, the WhatsApp number it is linked to will be used for bot replies. Make sure you use a number dedicated to the bot, or a WhatsApp Business account on a separate number.

3. Accounts and programs you need

What	Why	Cost	Link
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WhatsApp on your phone	To scan the QR code and link the bot	Free	whatsapp.com
Gmail account	The bot emails you the QR code image	Free	gmail.com
Node.js	Runs the bot on your computer	Free	nodejs.org
GitHub account	Stores your code, connects to Railway	Free	github.com
Railway account	Keeps the bot running 24/7 in the cloud	Free tier available	railway.app

1 Install Node.js on your computer

Node.js lets your computer run the bot. You only need to do this once.

1. Open your browser and go to **<https://nodejs.org>**
2. Click the big green button labelled **LTS** (Long Term Support)
3. Run the downloaded installer and click Next/Install with all defaults
4. When it finishes, **open a terminal**:
 - **Windows:** press `Win + R`, type `cmd`, press Enter
 - **Mac:** press `Cmd + Space`, type `Terminal`, press Enter
5. Type these two commands and press Enter after each:

terminal

```
node -v  
npm -v
```

✓ If both commands print a version number (e.g. `v22.0.0`), Node.js is installed correctly.

2 Get the project files

You need the bot's code files on your computer. There are two ways:

Option A — Download as ZIP (easiest)

1. Go to your GitHub repository in the browser
2. Click the green **Code** button
3. Click **Download ZIP**
4. Extract (unzip) the downloaded file to a folder like `C:\bots\my-whatsapp-bot`

Option B — Clone with Git

1. Install Git from <https://git-scm.com> if not already installed
2. Open a terminal and run:

terminal

```
git clone https://github.com/YOUR_USERNAME/my-whatsapp-bot.git
cd my-whatsapp-bot
```

3 Install the bot packages

The bot needs several helper libraries. Install them by running one command inside the project folder.

1. Open a terminal
2. Navigate to your project folder:

terminal – Windows

```
cd C:\bots\my-whatsapp-bot
```

terminal – Mac / Linux

```
cd /Users/yourname/bots/my-whatsapp-bot
```

Then run:

terminal

```
npm install
```

✓ You will see a lot of text scrolling. It finishes with a line like *"added 200 packages"*. That is normal.

4 Set up Gmail App Password

The bot emails you the WhatsApp login QR code. To do this securely, you create a special one-time password for the bot inside your Google account.

1. Open a browser and go to **<https://myaccount.google.com>**
2. Sign in to the Gmail account you want to use
3. Click **Security** in the left panel
4. Find **2-Step Verification** and turn it on if not already on
 - Follow Google's steps — it takes about 2 minutes
 - You will need your phone to receive a code
5. After turning on 2-Step Verification, search for **App passwords** in the Google search bar at the top of the page
6. Click on **App passwords**
7. In the *App name* field, type: `WhatsApp Bot`
8. Click **Create**
9. Google will show you a 16-character password like: `abcd efgh ijkl mnop`
10. **Copy this password and save it somewhere safe** — you will only see it once

 This app password is not your normal Gmail password. It is a special password for the bot only. Keep it private.

You will use:

- Your Gmail address as `EMAIL_USER` (e.g. `yourname@gmail.com`)
- The app password as `EMAIL_PASS` (e.g. `abcd efgh ijkl mnop`)

5 Find your WhatsApp Admin ID

The admin ID tells the bot who can send it a new product CSV. It is your WhatsApp phone number in a special format.

Format rules:

- Use your full international phone number
- Do **not** include the **+** sign
- Do **not** include any spaces or dashes
- Add **@s.whatsapp.net** at the end

Your phone number	Your ADMIN_JID
+27 82 123 4567	27821234567@s.whatsapp.net
+44 7700 900123	447700900123@s.whatsapp.net
+1 555 123 4567	15551234567@s.whatsapp.net

The phone number must be the same WhatsApp number you will use to scan the QR code and link the bot.

6 Set up your environment variables

Environment variables store your private settings outside the code. The bot reads three of them:

Variable	What to put in it	Example
<code>ADMIN_JID</code>	Your WhatsApp number (Step 5 format)	<code>27821234567@s.whatsapp.net</code>
<code>EMAIL_USER</code>	Your Gmail address	<code>yourname@gmail.com</code>
<code>EMAIL_PASS</code>	Your Gmail app password (Step 4)	<code>abcd efgh ijkl mnop</code>

Create a file called `.env` in your project folder (no extension) and add these lines:

`.env` file

```
ADMIN_JID=27821234567@s.whatsapp.net
EMAIL_USER=yourname@gmail.com
EMAIL_PASS=abcd efgh ijkl mnop
```

⚠ Never share this file or commit it to GitHub. It contains your private password. The `.gitignore` file in the project blocks it from being uploaded automatically.

7 Start the bot on your computer

If you are using a .env file

Install the dotenv package first:

terminal

```
npm install dotenv
```

Then add this as the very first line of `index.js` :

index.js – first line

```
require('dotenv').config();
```

Then start:

terminal

```
npm start
```

Mac or Linux (without .env file)

terminal

```
ADMIN_JID="27821234567@s.whatsapp.net" \  
EMAIL_USER="yourname@gmail.com" \  
EMAIL_PASS="abcd efgh ijkl mnop" \  
npm start
```

Windows PowerShell (without .env file)

PowerShell

```
$env:ADMIN_JID="27821234567@s.whatsapp.net"  
$env:EMAIL_USER="yourname@gmail.com"  
$env:EMAIL_PASS="abcd efgh ijkl mnop"  
npm start
```

 You should see:

Health check server listening on port 3000



Initializing WhatsApp Engine...



Inventory Loaded



QR Code generated. Sending to email...

8 Scan the QR code on your phone

The bot cannot talk to WhatsApp until you link it to your phone number. You do this once by scanning a QR code.

1. After starting the bot, wait up to 2 minutes for a QR code email to arrive in your Gmail inbox
2. If it does not arrive, check the **spam/junk** folder
3. Open the email — it has an attached image called `bot-qr.png`
4. On your phone, open **WhatsApp**
5. Tap the three dots (:) in the top right corner
6. Tap **Linked Devices**
7. Tap **Link a Device**
8. Point your phone camera at the QR code image on your computer screen
9. Wait 5–10 seconds

✓ In the terminal you will see:

🚀 BOT IS CONNECTED AND LIVE!

QR code expires after about 60 seconds. If the bot sends another QR code email, just scan the newest one.

WhatsApp Business tip: Using a WhatsApp Business account on a dedicated number is recommended so your personal messages stay separate.

9 Test the bot

From a **different** WhatsApp number, send these messages to the bot's number:

Message to send	Expected reply
hello	The full product list
menu	The full product list
buy 1 2	"✅ Added 2 x Coffee Beans 500g."
buy 3 1	"✅ Added 1 x Espresso Machine."
checkout	Order summary and total price

✅ If the bot replies to all of these, it is working correctly.

10 Manage your products

Your product list is stored in `products.csv` in the project folder. Edit it with Excel, Google Sheets, or any text editor.

File format

`products.csv`

```
ID,Name,Price
1,Coffee Beans 500g,15.00
2,Coffee Filters,5.50
3,Espresso Machine,299.00
```

Rules:

- The first line must always be exactly: `ID,Name,Price`
- Each product needs a unique ID number
- Price should be a plain number only (no currency symbol)
- Do not add extra columns or empty rows

Update products without restarting

From the admin WhatsApp number, send the updated `products.csv` file as a document attachment directly to the bot's number. The bot will automatically load the new products and reply "📦 *Products updated!*"

How to deploy on Railway (always-on cloud)

Running the bot on your computer means it stops when you close the terminal. Railway is a free cloud hosting service that keeps the bot running 24/7.

A Create a GitHub account and push your code

1. Go to <https://github.com> and create a free account
2. Create a new repository called `my-whatsapp-bot`
3. Push your project files to it:

terminal

```
git init
git add .
git commit -m "Initial bot setup"
git remote add origin https://github.com/YOUR_USERNAME/my-whatsapp-bot.g
git push -u origin main
```

The `.gitignore` file in the project automatically blocks `.env` and `node_modules/` from being uploaded to GitHub.

B Create a Railway account and project

1. Go to <https://railway.app>
2. Click **Login** and sign in with your GitHub account
3. Click **New Project**
4. Choose **Deploy from GitHub repo**
5. Select your `my-whatsapp-bot` repository
6. Railway will detect the project and prepare a deployment automatically

C Add environment variables to Railway

In Railway you set your private settings in the Variables tab — **never in the code itself**.

1. In your Railway project, click on your service
2. Click the **Variables** tab
3. Click **New Variable** and add each one:

Variable name	Value
ADMIN_JID	e.g. <code>27821234567@s.whatsapp.net</code>
EMAIL_USER	e.g. <code>yourname@gmail.com</code>
EMAIL_PASS	your 16-character Gmail app password

✓ Railway will automatically restart the service with these variables loaded.

D Link WhatsApp after Railway deploys

1. In Railway, click on your service and open the **Logs** tab
2. Watch the logs — you will see `Initializing WhatsApp Engine...`
3. Check your Gmail inbox for the QR code email
4. Scan the QR code with WhatsApp on your phone (same as Step 8)
5. Logs will show `🚀 BOT IS CONNECTED AND LIVE!`

Railway keeps the bot running automatically. If it restarts, it reconnects to WhatsApp using saved session files without needing a new QR scan — unless WhatsApp logs it out.

Troubleshooting

Problem	Solution
Missing required environment variables	You forgot to set ADMIN_JID, EMAIL_USER, or EMAIL_PASS. Check your .env file or how you started the bot.
QR code email never arrives	1. Check Gmail spam folder 2. Make sure you used a Gmail address (not Outlook/Yahoo) 3. Check that the app password is the 16-character one from Step 4 4. Make sure 2-Step Verification is turned on in Google
QR code expired before I could scan it	A new QR code will be emailed automatically. Just scan the latest one.
Bot replies "Cart empty" at checkout	Add items first using <code>buy [id] [qty]</code> before using checkout.
Bot shows no products or wrong products	Check that <code>products.csv</code> is in the project root folder and the first line is exactly <code>ID,Name,Price</code> .
Bot does not respond to messages	Make sure the bot is running and you see  <code>BOT IS CONNECTED AND LIVE!</code> in the logs. Also check you are messaging the correct WhatsApp number.
Railway deploy fails	1. Check all three variables are set in Railway's Variables tab 2. Check Railway logs for the specific error 3. Make sure <code>package.json</code> has a <code>start</code> script
WhatsApp logged the bot out	The bot deletes session files and generates a new QR code automatically. Check Gmail for a new QR code email and scan it.
<code>npm install</code> shows vulnerability warnings	These come from the WhatsApp library and are pre-existing. The bot still runs normally.

Quick Command Reference

Customer commands (send these to the bot on WhatsApp)

```
# Show the product menu
hello    or    menu

# Add items to basket: buy [product ID] [quantity]
buy 1 2 → adds 2 of product ID 1
buy 3 1 → adds 1 of product ID 3

# See order total and clear the basket
checkout
```

Admin command (from your WhatsApp number only)

```
# Send a new products.csv file as a WhatsApp document attachment
→ Bot replies: Products updated!
```

Terminal commands

```
# Install packages (run once after downloading the project)
npm install

# Start the bot
npm start

# Check Node.js version
node -v
```

